

Vihanga Chamodya Kumanayaka

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Summary

I am a 3rd-year undergraduate at the University of Moratuwa with experience and passion in mechatronic system designing and related sectors. My technical experiences extend to embedded systems, simulations and modelling, programming, electronic circuit designing, CAD modelling and FEA. My involvement in extracurricular activities have sharpened my skills, such as team leadership, project coordination, conflict resolution, time management, risk management and communication. I am passionate about using my engineering, social and intellectual skills to explore interesting areas in engineering and to create engineering solutions to overcome demanding real-world problems.

Education

**B.Sc Engineering (Hons) in Mechanical Engineering,
Specializing in Mechatronic Systems Engineering**
Current GPA: 3.69/4.00

University of Moratuwa, Sri Lanka

May 2024 - Present

G.C.E. Advanced Level Examination
Physical Science Stream
Z-Score – 2.2173

Sri Sumangala College, Panadura, Sri Lanka

August 2020 - January 2023

G.C.E. Ordinary Level Examination
9A's

Sri Sumangala College, Panadura, Sri Lanka

January 2018 - December 2019

Projects

Mechatronic System Design Project

Ongoing

Otter morphology-mimicking robot for aquatic environment exploration. A mechatronic system inspired by the biomechanisms of an otter's swimming pattern. Current work focuses on designing a body linkage system for undulatory motion of the spine and tail, and a linkage system for paddling of otter limbs.

Micromouse Robot

Ongoing

Microcontroller programming, sensor integration and communication with MCU, motor control, sensor fusion using an Extended Kalman Filter, implementing maze-solving algorithms and SLAM, CAD design.

Productions and Operations Management Project

Ongoing

Visiting an industrial organisation in the production sector, analysing current production methods and operations management methods, and proposing improvements for areas where changes can be made. Current work includes mapping out the factory floor layout and worker foot traffic.

DIYAKAWA 3.0 - AUV Project

December 2025 - April 2026

Designed an AUV for the SAUVC 2026 competition. My contributions include designing and implementing electronic circuitry, programming robot logic, and designing sealing methods for underwater compartments.

Reverse Engineering Project

January 2026 - April 2026

Reverse-engineered an IR cooker. My work includes identifying the internal ceramic housing material, researching material properties, researching manufacturing methods, and CAD modelling of the component.

Instrumentation System Design Project

July 2025 - November 2025

Designing and implementing a system to track metrics of a bicycle. My work includes implementing an IMU (accelerometer and gyroscope) and sensor fusion using an Extended Kalman Filter.

Mechatronic System Engineering Project

January 2025 - May 2025

Developing a SCARA robot for automated object picking and placing. My work includes integrating a Raspberry Pi 4 with a YOLOv8 model for the computer vision system.

Manufacturing Project

January 2025 - May 2025

Using woodwork, sheet metal forming, turning, and welding operations to fabricate a vintage movie projector replica.

Work/Volunteering experience

Falcon E Racing Team – University of Moratuwa Formula Student Racing Team

Vehicle Dynamics Subsystem Member

December 2025 - July 2026

- Analyzed vehicle dynamic performance using a vehicle model in Matlab Simscape Multibody.
- Fit tire testing data to the Pacejka Formula and derived Pacejka coefficients.
- Designed an IR sensor-based temperature sensor array for tire temperature measurement.

Extracurricular activities

MoraSpirit Initiative – Corporate Development Pillar

July 2024 - Present

- Worked in cold-calling teams for Inter University Games media coverage 2024, SpiritX 2024, and SLUG 2025.
- Co-chaired SpiritFlicks 2.0.

Member of University Karate Team

May 2024 - Present

- Reserve player for men's below 55 kg Kumite in the Inter-University Tournament 2024.
- Quarter-finalist in men's below 55 kg Kumite, SLUG 2025.
- District 2nd place in men's below 60 kg Kumite National Sports Festival Karate Tournament, 2026.
- Received University of Moratuwa Colours awards for years 2024 and 2025.
- Represented the University of Moratuwa in numerous karate tournaments.

Technical skills

Programming Languages	Python, C, C++, MATLAB, Assembly
Frameworks	ROS 2, YOLOv8, OpenCV
Embedded Systems/Microcomputers	Raspberry Pi, ESP32, Arduino
Software	CAD (Solidworks, Fusion360, AutoCad), PCB designing (KiCAD, Proteus), FEA (Ansys, COMSOL Multiphysics), MBD (Adams), Simulations and modelling (MATLAB), Embedded System Programming (Atmel Studio), Pneumatic system designing (Fluid-Sim)
Computer Skills	MS Office, Canva

Language proficiencies

English
Sinhala